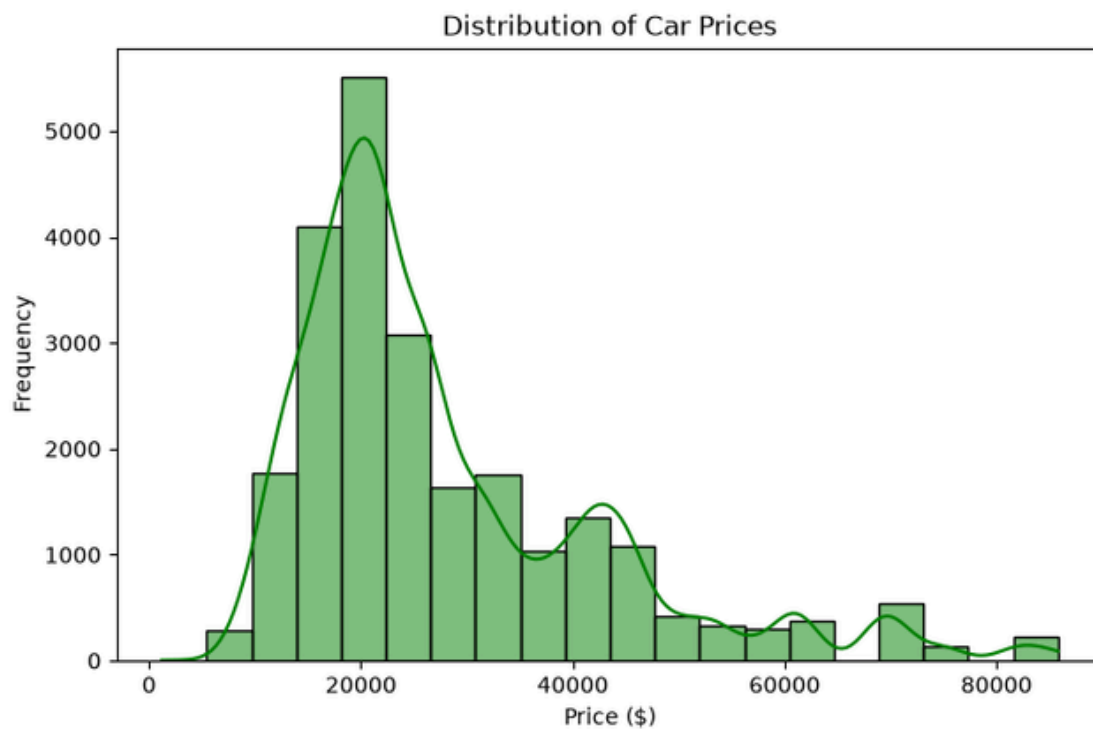
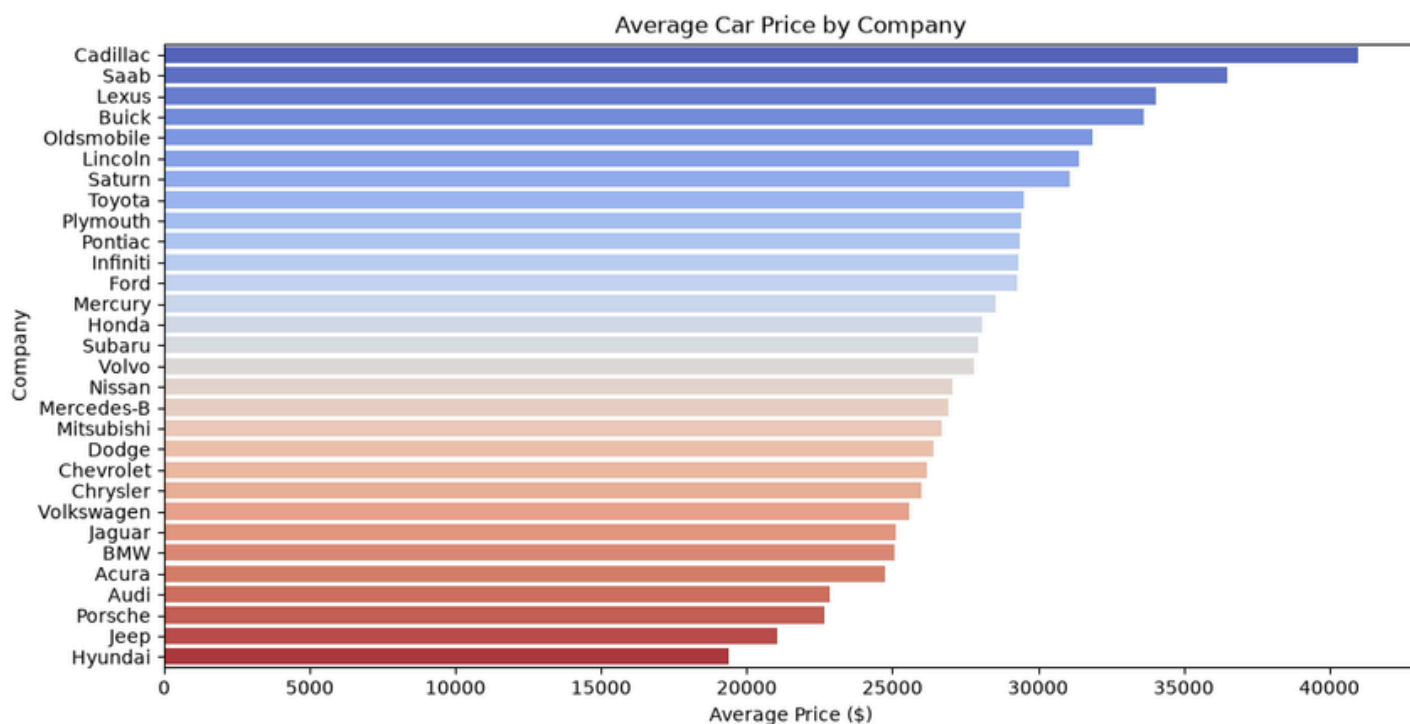


```

1  import pandas as pd
2  import seaborn as sns
3  import matplotlib.pyplot as plt
4
5  df = pd.read_excel('Car Sales Dataset.xlsx')
6  df
7
8  #Histogram Bar Price Distribution
9  plt.figure(figsize=(8,5))
10
11  sns.histplot(df['Price ($)'],bins=20,kde=True,color='green')
12  plt.title('Distribution of Car Prices')
13  plt.xlabel('Price ($)')
14  plt.ylabel('Frequency')
15  plt.show()
16
17
18
19  # Bar Chart Average Price by Company
20  plt.figure(figsize=(12,6))
21
22  avg_price = df.groupby('Company')['Price ($)'].mean().sort_values(ascending=False)
23
24  sns.barplot(x=avg_price.values,y=avg_price.index,palette='coolwarm')
25
26  plt.title('Average Car Price by Company')
27  plt.xlabel('Average Price ($)')
28  plt.ylabel('Company')
29  plt.show()
30
31

```

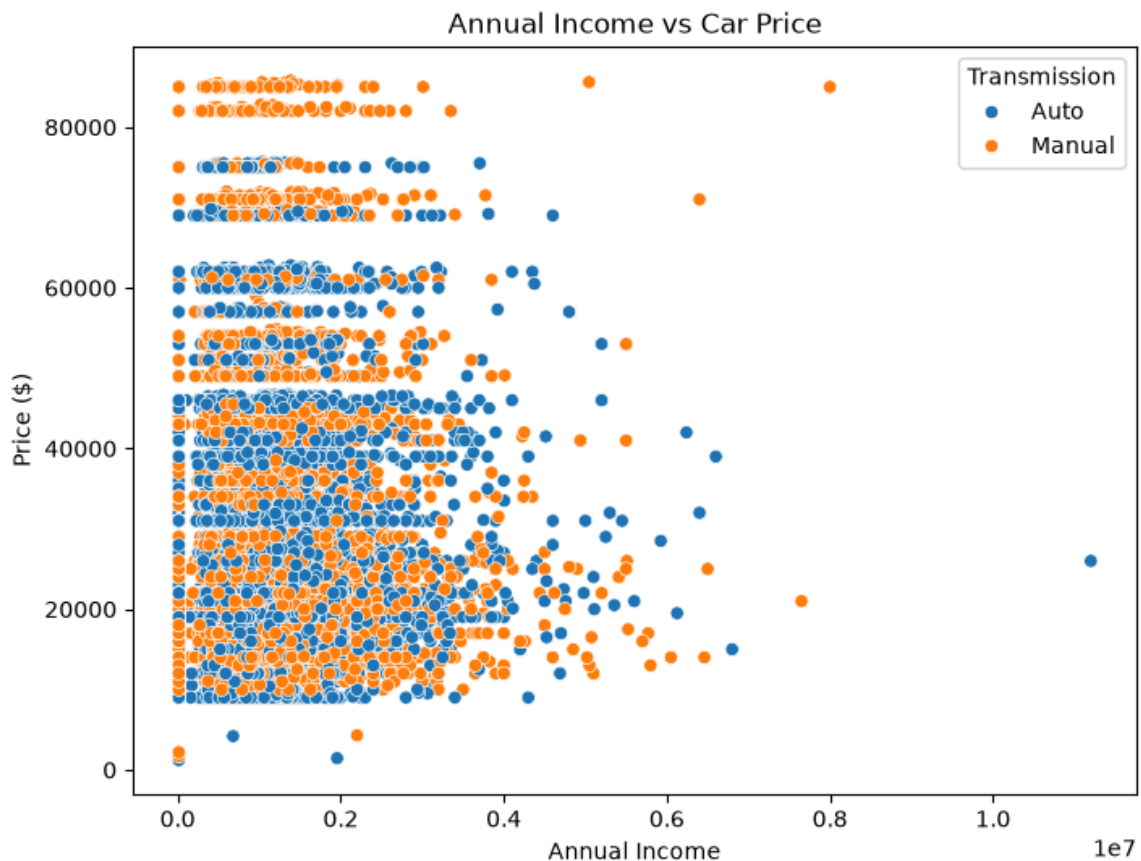


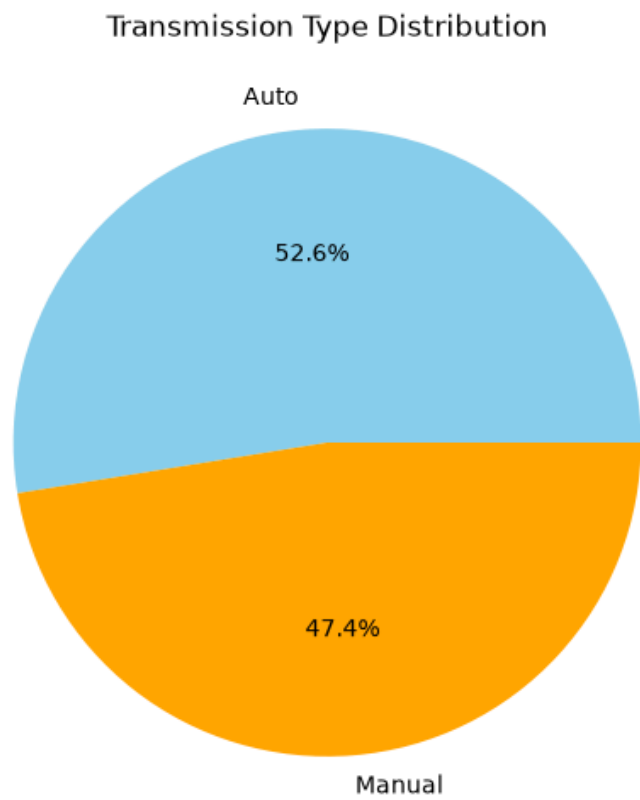
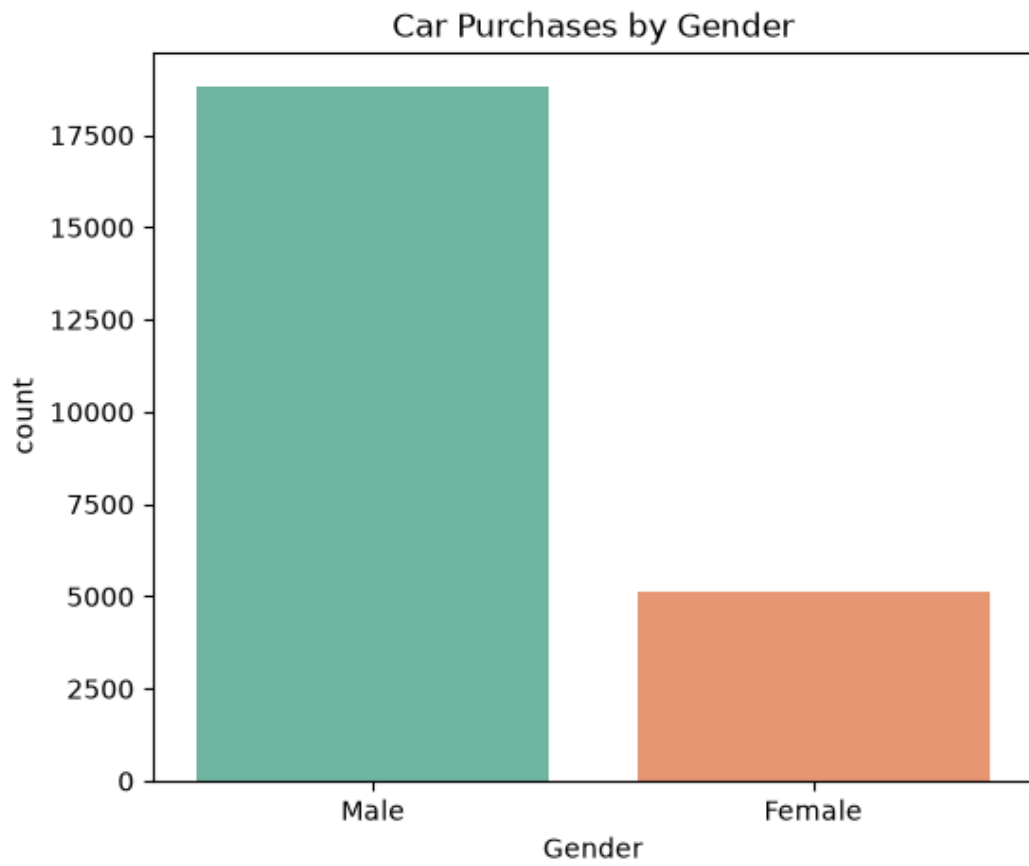


```

32 #scatter plot Annual Income vs Car Price
33 plt.figure(figsize=(8,6))
34
35 sns.scatterplot(data=df,x='Annual Income',y='Price ($)',hue='Transmission')
36
37 plt.title('Annual Income vs Car Price')
38 plt.show()
39
40
41 #count Car Purchases by Gender
42 plt.figure(figsize=(6,5))
43
44 sns.countplot(data=df,x='Gender',palette='Set2')
45
46 plt.title('Car Purchases by Gender')
47 plt.show()
48
49
50 #Transmission Distribution (Pie Chart)
51 df['Transmission'].value_counts().plot(kind='pie',autopct='%1.1f%%',figsize=(6,6),colors=['skyblue','orange'])
52 plt.title('Transmission Type Distribution')
53 plt.ylabel('')
54 plt.show()
55
56
57 #Box Plot of Price by Transmission
58 plt.figure(figsize=(8,5))
59 sns.boxplot(data=df,x='Transmission',y='Price ($)',palette='pastel')
60
61 plt.title('Price Distribution by Transmission')
62 plt.show()
63

```





```

56
57 #Box Plot of Price by Transmission
58 plt.figure(figsize=(8,5))
59 sns.boxplot(data=df,x='Transmission',y='Price ($)',palette='pastel')
60
61 plt.title('Price Distribution by Transmission')
62 plt.show()
63
64
65
66 #Correlation Heatmap relationship between income and price
67 numeric = df[['Annual Income','Price ($)']]
68
69 plt.figure(figsize=(6,4))
70
71 sns.heatmap(numeric.corr(),annot=True,cmap='coolwarm',linewidths=0.5)
72
73 plt.title('Correlation Heatmap')
74 plt.show()

```

